

# Abdul Haque

AI Software Engineer | Backend Systems, LLMs & Agentic AI

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## Summary

Backend AI Engineer specialising in production LLM microservices (FastAPI) and multi-agent RAG systems (LangGraph, LangChain, CrewAI) on Azure, with hands-on experience deploying a CPU-only quantised inference pipeline serving **300+ daily sessions** for an Australian government healthcare client.

## Technical Skills

**Programming:** Python, SQL, Bash, C++

**Backend:** FastAPI, REST APIs, microservices, AsyncIO, Pydantic, OAuth/JWT, Middleware

**Cloud & Infra:** Azure (AKS, DevOps, Blob Storage), AWS (boto3), GCP, Docker

**AI/LLM:** LangGraph, LangChain, CrewAI, multi-agent systems, OpenAI/Anthropic APIs, Azure OpenAI, vLLM, Ollama

**ML/NLP & Speech:** PyTorch, HuggingFace Transformers, Scikit-learn, Faster-Whisper

**Databases & RAG:** PostgreSQL (pgvector), Neo4j (Cypher), GraphRAG

**Messaging & Observability:** Azure Service Bus, Kafka, MLflow, Prometheus, Grafana

## Professional Experience

### Associate Software Engineer – AI

Dec 2025 – Present

*DeltaShoppe (Pvt.) Ltd, Islamabad | Partner: Kermit Tech (Norway) Sole backend engineer rotating across three concurrent client engagements within a small AI engineering pod.*

#### AI Therapy Microservice

*Azure AKS, Faster-Whisper, Phi-3.5-Mini, Azure Service Bus, FastAPI*

- Designed AI therapy microservice for **Australian government healthcare client**; **300+ daily sessions** with event-driven audio pipeline on Azure Service Bus decoupling transcription/summarisation into background jobs.
- Deployed **Faster-Whisper + Phi-3.5-Mini (INT8)** CPU-only on 15 GB AKS pods; **multi-processing worker pool** resolved lock-contention and ONNX OOM failures.
- JWKS token verification, encrypted audio at rest, dead-letter queues with retry logic, **MLflow** observability.

#### AI Student Companion

*LangGraph, FastAPI, PostgreSQL (pgvector), Docker*

- Built **multi-tenant RAG system** with pgvector, per-client isolation, and propositional chunking (~10:1 compression, **90% token reduction**). Sliding-window memory with importance scoring cut tokens from 100K to 10K/session.
- Tool injector pattern for per-user permission scoping across LLM tools; PII redaction middleware and audit logging.
- Deployed via **Azure DevOps CI/CD**; retry/fallback across Groq/Mistral; load-tested at 5–10 concurrent sessions.

#### Real-Time Fraud Detection & Regulatory Reporting

*Kafka, StreamSets, PostgreSQL, Docker, FastAPI*

- Built real-time fraud detection for **regulated retail banking**: event-driven pipeline on Kafka with PII masking (6 fields) and automated **Suspicious Transaction Report** filing to regulator every 5 min.
- Monitored via **Prometheus + Grafana** alerting (3 dashboards) with CRITICAL alerts on verification failures.

## Earlier Internships

- Data Science Intern, Securely Innovations (Pvt.) Ltd (Apr–Nov 2024)** – Built a multi-threaded data collection pipeline scraping ~3M automotive parts records via network-level API replication, outperforming Selenium-based scrapers by 3×.
- Computer Vision & Generative AI Intern, DataInsight Lab (FAST-NUCES) (Jun–Aug 2024)** – Fine-tuned Stable Diffusion (DreamBooth + LoRA, QLoRA) for furniture/interior design generation on 2×T4 GPUs using model parallelism.

## Research & Projects

### GraphRAG-Causal: Graph-Augmented Framework for Causal Reasoning in News

Aug 2024 – Apr 2025

*Stack: CrewAI Flows, FastAPI, Docker, Neo4j | [GraphRAG-Causal](#) | [arXiv](#)*

- Causal GraphRAG framework using XML cause-effect annotation + multi-hop graph traversal; **82.88% F1 / 80% accuracy** on CausalNewsCorpus, on par with fine-tuned BERT-Large.
- Agentic extension in CrewAI Flows integrating graph DB and web search for controllable causal inference.

### IntentLens: Multi-Agent Feedback Analysis

Aug 2024 – Jan 2025

*Stack: CrewAI, Groq API, Streamlit*

- Built a 4-agent system (scraper, intent analyser, search agent, supervisor) analysing 5,000+ customer reviews with LLM-as-judge evaluation.

## Education

**B.Sc. Data Science** – FAST-NUCES, Islamabad

Aug 2021 – Jun 2025

CGPA: 3.24/4.0 | Final-Year GPA: 3.75/4.0 | **Dean's List:** Fall 2024, Spring 2025 | FYP: GraphRAG-Causal

## Certifications & Leadership

**Generative AI with LLMs** – DeepLearning.AI & AWS | **Section Leader, Code in Place** (Stanford University, 2025) – 95% student retention across a global cohort.